

11.1 Exploring Exponential Growth Versus Decay

Lesson Introduction

 Benchmark	MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.		 Learning Targets	- I can classify a function as representing exponential growth or decay from a graph. - I can classify a function as representing exponential growth or decay from a table.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	- How can a function be classified as exponential growth or decay from a graph? - How can a function be classified as exponential growth or decay from a table?				
 Lesson Narrative	In this lesson, students are introduced to the concepts of exponential growth and exponential decay. The lesson is designed to be done through a Desmos activity or using the workbook. Students will begin by engaging in a collaborative activity where they examine real-world scenarios that model exponential growth and decay without learning the formal terminology. They are then formally introduced to the concepts as they continue connecting their exploration to graphs and tables. Students will bring everything together through learning about exponential equations.				
 Component Summary	11.1.1 Warm-Up (~5 min.)	Notice and Wonder activity on snake sighting data (<i>SE p. 155</i>)	11.1.4 Bring It Together (5-10 min.)	Introduction to exponential equations (<i>SE p. 159</i>)	
	11.1.2 Exploration (5-10 min.)	Explore real-world exponential growth and decay scenarios (<i>SE p. 156</i>)	11.1.5 Your Turn! (10 min.)	Practice with classifying exponential functions using tables and graphs (<i>SE p. 160</i>)	
	11.1.3 Collaboration (10-15 min.)	Collaborative activity with exponential tables and graphs (<i>SE p. 157</i>)	11.1.6 Wrap-Up (~5 min.)	Lesson reflection activity (<i>SE p. 162</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Exponential function - Exponential growth - Exponential decay - Exponential equation <u>Additional Terms:</u> N/A		(Optional) Graphing utility		- Preview the Desmos activity for 11.1.2 Exploration to determine whether to facilitate with that or use only the workbook. - The 11.1.2 Exploration contains optional videos (in the fourth and twenty-first slides of the Desmos activity) to accompany the questions prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may read the graph from the wrong direction, switching their classifications of growth and decay. - Students may struggle with determining growth or decay from a table of values.	- This is the first time students are being introduced to exponential functions. They have worked with exponents in both their prior course and earlier in this course. Connect what they have learned about exponents to the structure of expressions used in this lesson. - There are a few places within this lesson where students may get caught up in the nuances of the question, particularly with the real-world examples in 11.1.2 Exploration. Complete the activities on your own prior to facilitating with students to better understand where they may get stuck.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 11.1.1 Warm-Up has students review a visual data display of snake sighting data in or near Everglades National Park. Students will articulate what they notice about the graph and what they wonder about pythons found in or near Everglades National Park.	MA.K12.MTR.1.2: Participate in Effortful Learning This lesson is the first time that students are introduced to exponential growth and decay, which can be a challenge for students. 11.1.2 Exploration provides students with an exploration activity and is followed up with 11.1.3 Collaboration. Students should help and support each other when attempting a new method or approach in this lesson as they explore a new concept.
 Differentiate Instruction	11.1.3 Collaboration connects the students’ exploration in the prior activity to exponential graphs and tables. Consider pairing students up based on their dominant learning style to provide appropriate scaffolds where necessary that benefit pairings of students with similar strengths and preferences. For 11.1.3 Collaboration, consider creating the different-sized images for the students to manipulate and explore while they are completing the questions. 11.1.5 Your Turn! provides students with the opportunity to determine whether the representation provided is an exponential growth, decay, or neither, along with justification. Student performance on this activity can show which students may need more support in upcoming lessons as they develop their understanding of these types of functions.	
Lesson Notes:		

11.1.1 Warm-Up: Invasive Snakes

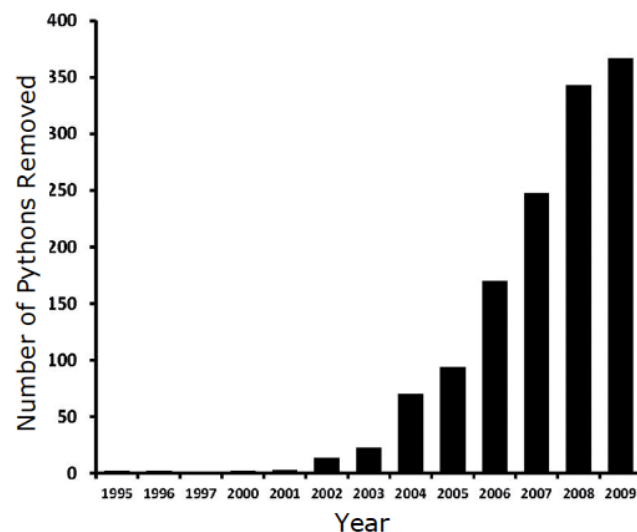
Component Overview: Students will review data on snake sightings in or near Everglades National Park. They will use this data to describe the graph, what they notice, and what they wonder.



11.1.1 Warm-Up: Invasive Snakes

1. The Burmese python is one of the largest species of snakes and is one of the most concerning invasive species in Everglades National Park. The graph below shows the number of pythons removed each year from 1995 to 2009.

Scientists now estimate the population of Burmese pythons in the Everglades is more than 100,000.



- a. What do you notice about the graph?
Answers will vary. After the year 2002 the graph continues to increase at a faster rate.
- b. Given the graph, are you surprised scientists now estimate more than 100,000 snakes in Everglades National Park?
Answers will vary. I am not surprised because of the rapid increase in the number removed.



In this activity, students are engaging in a Notice and Wonder routine. Allow the students a few minutes to independently answer the questions. After they have had a chance to write down their responses, ask several students to share things they noticed and wondered about the graph and its implications.

11.1.2 Exploration: Dominoes and Stickers

Component Overview: Students will explore the concepts of exponential growth and decay without a formal introduction to the terms, using two real-world scenarios: dominoes and stickers. This activity is designed to be done with a partner or small group.



11.1.2 Exploration: Dominoes and Stickers

1. Dr. Stephen Morris, a physics professor at the University of Toronto, set up a series of 13 dominoes, each roughly $1\frac{1}{2}$ times the size of the one knocking it over. The original domino is 5 millimeters (mm) tall and only 1 mm thick. The 13th domino is 3.2 feet (ft) tall and weighs about 100 pounds.



- a. What do you wonder about the dominoes?
Answers will vary. I wonder if the smaller dominoes will eventually knock over the larger ones at the end.
 - b. Do you think starting with the first domino, which is smaller than a Tic Tac, will result in the 13th domino falling?
Answers will vary. I don't think it will because the small ones don't weigh as much. I think it will because of the amount of dominoes.
2. In a 1983 edition of the American Journal of Physics, Lorne Whitehead wrote about his discovery that as a domino falls it could topple things larger than itself. He found that a domino is capable of knocking down a domino that is 50% larger than itself.
- a. Use the information about the 13th domino to determine the height, in feet (ft), for the 14th and 15th dominos.

Domino	Height (ft)
13 th	3.2
14 th	4.8
15 th	7.2



This scenario comes from an actual experiment done that was captured on film. Consider playing the videos (slides four and twenty-one in the Desmos activity) for students after they complete both questions 1a and 1b of this component.



Why do you think it is possible for the dominoes to fall the way that they do when the first one is so small?



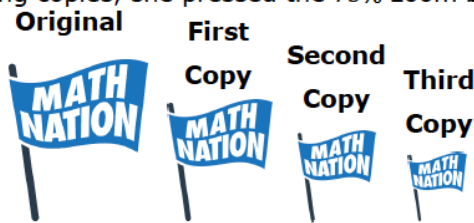
- Do you think that there could be a limit to how large a domino could be?
- In the video, Dr. Morris says that the 29th domino in the pattern would be as tall as the Empire State Building. How do you think that is possible based on this pattern?

11.1.2 Exploration: Dominoes and Stickers (continued)

- b. Discuss with your partner how the description that Dr. Morris uses, " $1\frac{1}{2}$ times itself," and the description Dr. Whitehead uses, "50% larger than itself," mean the same thing. Write your explanation or give an example to explain.

Answers will vary. Something being "50% larger than itself" would mean that it is itself (1) and 50% more (.5), making it 1.5 times larger, which is the same as " $1\frac{1}{2}$ times itself."

3. Ashley used a copy machine to make copies of a Math Nation sticker to observe how the size changes. Before making copies, she pressed the 75% zoom button.



- She made the first copy of the sticker.
- To make the second copy, she removed the original sticker from the copy machine and replaced it with the first copy of the sticker. She then pressed copy.
- To make the third copy, she removed the first copy of the sticker and replaced it with the second copy of the sticker. She then pressed copy.

How is each copy of the sticker related to the copy that was made before?

Each copy is 75% the size of the previous copy.



Create a space for students to share their thoughts on question 2b. It's important to ensure students understand this language prior to moving on to the remainder of the lesson.



Consider doing this activity on your own. Follow the exact same instructions that the teacher in this scenario used. Make multiple copies to provide as a physical manipulative for students to see as they explore question 3.



What is causing the flag to be smaller each time that it is copied?

11.1.3 Collaboration: Graphs and Tables

Component Overview: Students will be introduced to the formal definitions of exponential functions, exponential growth, and exponential decay. They will learn about the graph and table representations of both exponential growth and decay.



11.1.3 Collaboration: Graphs and Tables

There are two types of exponential functions: exponential growth and exponential decay.



An **exponential function** is a function with a constant percent rate of change.

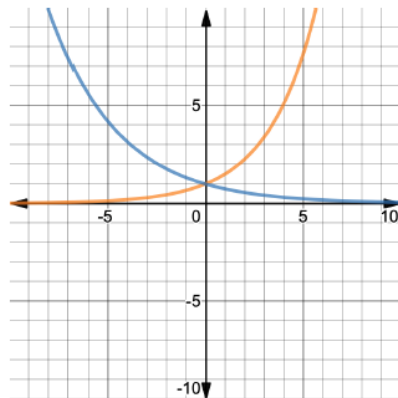


Exponential growth describes an exponential function whose constant percent rate of change is greater than 100%.



Exponential decay describes an exponential function whose constant percent rate of change is greater than 0% but less than 100%.

- Use the descriptions of exponential growth and decay to determine what the orange graph represents and what the blue graph represents.



Orange:

- Exponential growth
- Exponential decay

Blue:

- Exponential growth
- Exponential decay



Consider using a Frayer Model or anchor chart for these terms. It will be important for students to not only have the formal definition for these, but to have examples and nonexamples. Having this side by side on an anchor chart can be a helpful reference point for students to quickly recall the difference between the two.



Watch and listen for how students read the graph. You may find students that look at the orange graph and see a downward slope, or a “slide.” This can indicate that students aren’t reading the graph from left to right, but rather from right to left.

11.1.3 Collaboration: Graphs and Tables (continued)

2. One of the graphs models the size of the dominoes, and the other models the size of the sticker. Discuss with your partner how to determine which graph, orange or blue, is the function that models the size of the dominoes. Write a summary of your discussion.

Answers will vary. The dominoes example would be represented by the orange graph because the size of the dominoes continues to increase at a rapid rate. The sticker example would be represented by the blue graph because the size of the sticker continues to decrease at a rapid rate.

3. Each graph assumes the first sticker or the first domino is considered 100% of the original size.

What does the point $(0,1)$ represent for the function that represents the dominoes?

The point $(0,1)$ represents the original because 0 represents the beginning and 1 represents 100% because 100% written as a decimal is 1.00 or 1.



When reviewing the answers to this activity, allow a few extra minutes for students to share out their answers to question 2. Prompt students to explain how they know their answers are correct and what rationale they are using to arrive at that answer.

11.1.3 Collaboration: Graphs and Tables (continued)

4. The tables represent the two different contexts. Each table assumes that the first sticker or the first domino is 100% of the original size.

Table 1		Table 2	
x	y	x	y
0	1	0	1
1	$\frac{3}{2}$	1	$\frac{3}{4}$
2	$\frac{9}{4}$	2	$\frac{9}{16}$
3	$\frac{27}{8}$	3	$\frac{27}{64}$
4	$\frac{81}{16}$	4	$\frac{81}{256}$
5	$\frac{243}{32}$	5	$\frac{243}{1,024}$

- a. In Table 1, find the y -value for $x = 5$ using the y -value for $x = 4$. Then, explain how you determined your answer.

Answers will vary. In Table 1, I multiplied each y -value by $\frac{3}{2}$. In Table 2, I multiplied the prior y -value by $\frac{3}{4}$.

- b. Elise found that in Table 1, when she divided $\frac{9}{4}$ by $\frac{3}{2}$ or $\frac{27}{8}$ by $\frac{9}{4}$, she got the same value. How does the value she found relate to the context?

Answers will vary. It relates to the context because the dominoes are increasing by a constant factor.



Without calculating, what do we know would have to be true about the missing y -value in Table 1? What about Table 2? How do you know?



- *How can you find how one y -value changes to the next?*
- *How does the numerator change?*
- *How does the denominator change?*



For students who finish early, consider asking them to try and determine the missing value for when x is 10. Have them explain how they got their answer.

11.1.3 Collaboration: Graphs and Tables (continued)

- c. Use Elise's method and divide consecutive y -values in Table 2. How does the value relate to the context?

Answers will vary. The answer from dividing is $\frac{3}{4}$ which is equivalent to 75%. This value relates to the context because it is the scale factor by which Ashley was copying each sticker.

5. Discuss with your partner which table matches each context. Then, complete the statements based on your discussion.

Table 1 matches the ☐ stickers ☒ dominoes context because as

the x -values increase, the y -values in Table 1 are

☒ increasing ☐ decreasing at a constant percent rate

☐ between 0% and 100%.
☒ greater than 100%.

Table 2 matches the ☐ stickers ☒ dominoes context because as

the x -values increase, the y -values in Table 2 are

☐ increasing ☒ decreasing at a constant percent rate

☒ between 0% and 100%.
☐ greater than 100%.



Struggling readers may have a difficult time with how to approach this type of problem. This can be overcome by teaching students how to break the paragraph into individual sentences. Some students may find value in covering up the upcoming sentences with a sheet of paper to allow them to focus on each sentence individually.

11.1.4 Bring It Together: Exponential Equations

Component Overview: Students will connect what they did in the prior two components with exponential equations.



11.1.4 Bring it Together: Exponential Equations



An exponential equation is a type of equation in which the input is an exponent. It can be written in the form $y = a(b)^x$.

The a -value of an exponential equation is the initial amount.

From the previous contexts with the first sticker and first domino, those were considered 100% or 1. However, this is not the case for every situation. For example, the **dimensions** of the sticker or the **height** of the first domino could have been used as the initial value.

The b -value of an exponential equation is the growth or decay factor.

Exponential Growth	Exponential Decay
b is the growth factor	b is the decay factor
$b = 1 + r$, where r is the growth rate	$b = 1 - r$, where r is the decay rate
$y = a(1 + r)^x$	$y = a(1 - r)^x$

1. Complete the statement.

When the equation is written in the form $y = a(b)^x$, the

b is greater than 1 when the equation is modeling

exponential ☐ decay ☒ growth, and is between 0 and 1 when

the equation is modeling exponential ☐ decay. ☐ growth.



Understanding the structure of exponential equations is vital for success in the remainder of this unit. Consider creating an anchor chart that begins with this concept that you can add to later, as the key features of exponential functions are explored throughout the unit.



How is it possible to determine when an exponential function is growing or decaying based on the values in the equation?

11.1.4 Bring It Together: Exponential Equations (continued)

2. The equation that models the dominoes situation is $y = 1\left(1 + \frac{1}{2}\right)^x$ or $y = \left(\frac{3}{2}\right)^x$. The equation that models the sticker situation is $y = 1\left(1 - \frac{1}{4}\right)^x$ or $y = \left(\frac{3}{4}\right)^x$.

Discuss with your partner how the value of b in each equation relates to the relationship observed in the tables.

Answers will vary. The dominoes situation represented getting larger at a rapid rate, so it matches the growth equation. The sticker situation represented getting smaller at a rapid rate, so it matches the decay equation.



Allow students the opportunity to share their answers to question 2. This can create a great opportunity to build the conversation based on their understanding of the relationship between the different representations of exponential functions at this point in the lesson.

11.1.5 Your Turn!

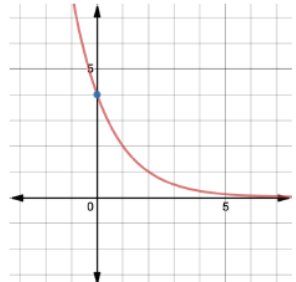
Component Overview: Students will determine whether a given representation of an exponential function represents decay, growth, or neither, with justification.



11.1.5 Your Turn!

1. Determine if each representation of a function is an example of exponential decay or exponential growth. Justify your selection.

a.

Representation	Type of Exponential Function
	☐ Exponential decay ☐ Exponential growth
Justification	
Answers will vary. The graph starts at (0,4) and gets closer to the x-axis at a rapid rate.	

b.

Representation	Type of Exponential Function
$y = 50(1 + 0.85)^x$	☐ Exponential decay ☐ Exponential growth
Justification	
The value of b is greater than 1.	



Consider using informal observation throughout the lesson to determine whether students would benefit from making this component more guided. This could mean small-group instruction for some, while others attempt independently, or it could mean teacher facilitation of one or two examples with students actively working as a whole group before releasing the class to complete the rest independently.



Consider having students leverage a Think-Pair-Share routine with this activity. Allow them to try these on their own and then compare their answer with a partner to check their work and come to consensus on the correct answer.

11.1.5 Your Turn! (continued)

c.

Representation	Type of Exponential Function												
<table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr><td>-3</td><td>4</td></tr> <tr><td>-2</td><td>8</td></tr> <tr><td>-1</td><td>16</td></tr> <tr><td>0</td><td>32</td></tr> <tr><td>1</td><td>64</td></tr> </tbody> </table>	x	y	-3	4	-2	8	-1	16	0	32	1	64	- Exponential decay - Exponential growth
x	y												
-3	4												
-2	8												
-1	16												
0	32												
1	64												
Justification													
<i>The y-values are increasing by a factor of 2.</i>													

d.

Representation	Type of Exponential Function
$y = 50(0.85)^x$	- Exponential decay - Exponential growth
Justification	
<i>The value of b is less than 1.</i>	

e.

Representation	Type of Exponential Function												
<table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr><td>0</td><td>$\frac{9}{2}$</td></tr> <tr><td>1</td><td>$\frac{9}{4}$</td></tr> <tr><td>2</td><td>$\frac{9}{8}$</td></tr> <tr><td>3</td><td>$\frac{9}{16}$</td></tr> <tr><td>4</td><td>$\frac{9}{32}$</td></tr> </tbody> </table>	x	y	0	$\frac{9}{2}$	1	$\frac{9}{4}$	2	$\frac{9}{8}$	3	$\frac{9}{16}$	4	$\frac{9}{32}$	- Exponential decay - Exponential growth
x	y												
0	$\frac{9}{2}$												
1	$\frac{9}{4}$												
2	$\frac{9}{8}$												
3	$\frac{9}{16}$												
4	$\frac{9}{32}$												
Justification													
<i>The y-values are decreasing at a rate of .50, which is less than 1.</i>													



Consider providing students with the opportunity to share out their answers in a whole-class review of the questions. There are some of these that can be trickier than others, and it will be important for students to hear from their peers as part of the learning process.



- Which types of representations were the easiest to determine the type of exponential function?
- Which types of representations proved to be the most challenging when determining the type of exponential function?

11.1.6 Wrap-Up: Reflection

Component Overview: Students will complete a reflection about their learning from this lesson. This activity is designed to be done independently to gather student-level data.



11.1.6 Wrap-Up: Reflection

- 1. Complete the table.
Answers will vary.

The terms and information used in the lesson that I knew were ...	Something I wondered during the lesson was ...	Something I learned during the lesson was ...



The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.